Library Management System

User Manual

Team Members

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1. Starting the application

Docker setup

```
docker compose up -d --build
docker compose exec app npm run seed
```

Open http://localhost:3000. Sign in with admin@library.edu and admin123. The librarian demonstration account uses librarian@library.edu with the same password.

Local setup

```
cp .env.example .env
npm install
npm run seed
npm start
```

2. Overview and navigation

The Overview page displays catalog titles, active members, issued books, overdue loans, reservation queues and recorded fines. The current-loan list highlights overdue records. On mobile screens, use the Menu button to open navigation.

Navigation item	Use
Overview	Daily status and inventory chart
Books	Catalog search and maintenance
Members	Member registration and maintenance
Circulation	Issue and return workflows
Reservations	Ordered hold queue
Reports	Six MongoDB analytics reports

3. Managing books

- Open Books and search by title, ISBN or publisher.
- Select Add book and enter a Bxxx ID, unique ISBN, author IDs and copy counts.
- Select Edit to change metadata or stock.
- Select Issue when a copy is available.
- Select Reserve when no copy is available.

Copy rule

Available copies cannot exceed total copies. Total copies cannot be reduced below the number currently issued.

4. Managing members

- Open Members and search by name, email or department.
- Select Register member and enter an Mxxx ID, contact details and membership type.
- Use Edit to update details or deactivate an account.
- Inactive members cannot issue or reserve books.
- A member cannot be deleted while an active loan exists.

5. Issue, return and reserve

Issue

Select Issue book, choose an available title, choose an active member and select the loan period. The application creates the loan and decreases stock in one transaction.

Return

Open Circulation and select Return beside an active loan. The application calculates Rs. 10 per overdue day, marks the loan returned, restores stock and fulfils the oldest reservation in one transaction.

Reserve

Reservations are allowed only when no copy is available. The queue is ordered by reservation date. Staff can cancel an active reservation.

6. Reports

Report	Interpretation
Most borrowed books	Titles with the greatest loan count
Inventory summary	Title and copy totals by category
Overdue loans	Member, overdue days and calculated fine
Member activity	Total loans, active loans and recorded fines
Monthly circulation	Loans and returns grouped by month
Reservation queues	Queue size and next member by book

Use Export overdue CSV to download the overdue list for evidence or follow-up.

7. Troubleshooting

Problem	Action
Login fails after a new database	Run npm run seed
Transaction error	Confirm MongoDB runs as a replica set
Database connection error	Check MONGODB_URI or docker compose ps
Port 3000 is busy	Change PORT in .env
No records appear	Run the seed command and refresh
Docker is paused	Resume Docker Desktop before docker compose commands

Security note

Change the seeded passwords and JWT secret before any non-classroom deployment.